



**DEPARTMENT OF EDUCATION
SCHOOLS DIVISION OF NEGROS ORIENTAL
REGION VII**

Kagawasan Ave., Daro, Dumaguete City, Negros Oriental



FORMATION AND SYNTHESIS OF HEAVIER ELEMENTS

Physical Science/ Grade 11 (Core Subject)

Quarter 3 / Week 1



SELF-LEARNING KIT

FOREWORD

This Self-Learning Kit will serve as a guide to give evidence for and describe the formation of heavier elements during star formation and evolution.

The concept of how atomic number led to the synthesis of new elements in the laboratory is also explained in this SLK.

It is aimed that through this Self-Learning Kit the learners will gain knowledge on the Formation and Synthesis of heavier elements.

OBJECTIVES

This self-learning kit aims to:

- a. give evidence for and describe the formation of heavier elements during star formation and evolution;
- b. demonstrate understanding of how the concept of atomic number led to the synthesis of new elements in the laboratory; and
- c. manifest awareness about the significance of these synthesized elements in medicine or saving lives of the people.

LEARNING COMPETENCIES:

Give evidence for and describe the formation of heavier elements during star formation and evolution (S11/12PS-111a-11)

Explain how the concept of atomic number led to the synthesis of new elements in the laboratory (S11/12PS-111b11)

I. WHAT HAPPENED

PRE-ACTIVITY

Part I. Directions: Read and understand the following statement and give your thoughts about it by writing in your activity notebook. Write at least 5 sentences.

"We are a way for the universe to know itself. Some part of our being knows this is where we came from. We long to return. And we can, because the cosmos is also within us. We're made of star stuff,"

Carl Sagan's **"The Cosmic Connection"**

Part II. Making Your Own Periodic Table

Directions: Make your own periodic table using the hypothetical elements that are given in the clues. Explain the word/s that will be formed if you arrange the symbols of the elements correctly. You can write this in your activity notebook. Do not write anything on this learning material.

- P** and **Pr** both have one electron each. **Pr** has a bigger atomic size.
- Od**, **Ri**, and **E** are in the same series as **P**, **C**, and **I**. In terms of atomic size, **P** is the biggest while **C** is the smallest. **E** is a metal while **I** is a non-metal. **Od** is smaller than **Ri** in atomic size.
- O** has a bigger atomic size than **E** in the same group. **Y** is also a bigger atom than **C** in the same group. **R** is more nonmetallic than **Pe** but more metallic than **Ti**.

I. WHAT I NEED TO KNOW

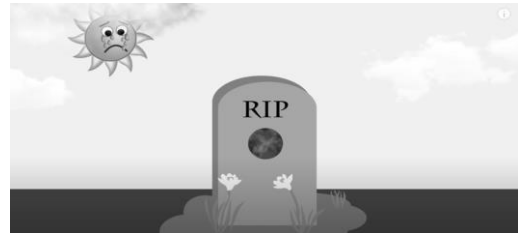
DISCUSSION:

Life of a Star

We've learned about what happened for the first billion years, or in the history of the universe which leaves us lots of stars and galaxies. We are now equipped with the terminology needed to describe and categorize these stars.

But we still have not talked about all other elements in the periodic table. we have only mentioned hydrogen and helium so far so where did the rest come from. And what about the planets and the moons? How did they get here?

The answer to this entire question will make sense once we learn more about what goes on inside the star, from the moment they are born, to the time of their death.



That's right, stars actually die, so to speak and the type of death along with what is left over, will be one of the variety of possibilities, depending entirely on the mass of a star.

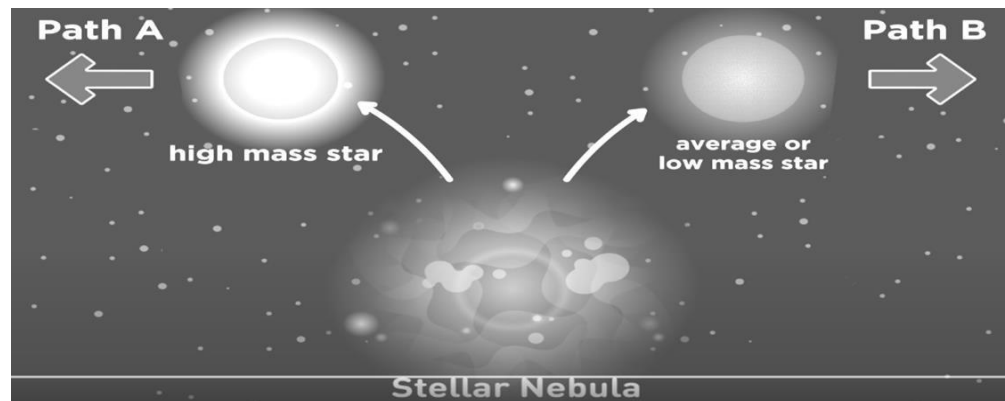
Let's go through the lifetime of a few different kind of stars, so that we are ready to understand the next 13 billion years of development of the universe.

Star Cycle

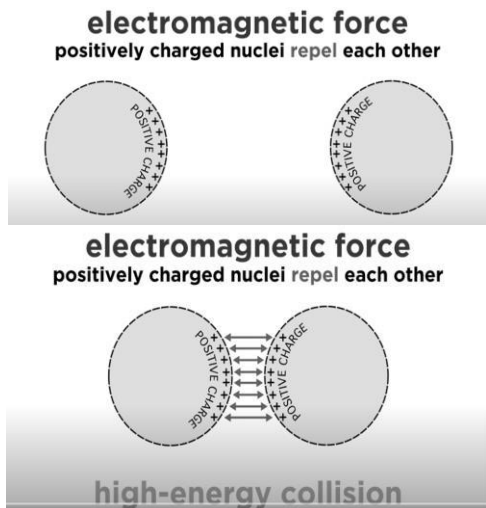
The life cycle of any star from birth to death, and all the stages in between, will span millions or billions of years. This is why

stars don't seem to change at all, because a human lifetime is a snippet of fraction of a blink of an eye to these giants.

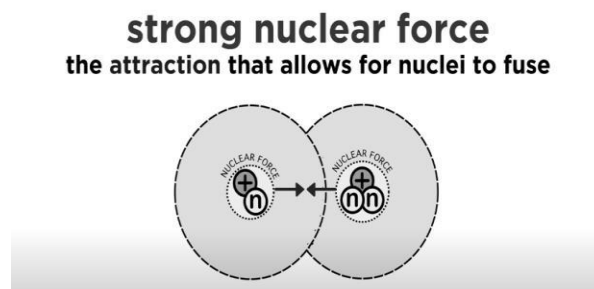
The path to be followed by a particular star depends mainly on its mass, or how much gas collected and collapsed to form a star, because that material will serve as a star's fuel.



<https://www.youtube.com/watch?v=4xIQGbYur9Q>



As we may remember from physics and chemistry, when nuclei collide with enough energy to overcome the electromagnetic repulsion between them, the stronger nuclear force take place, and they fuse small fraction of their mass converting into a huge amount of pure energy, as dictated by **E** equals mc squared.

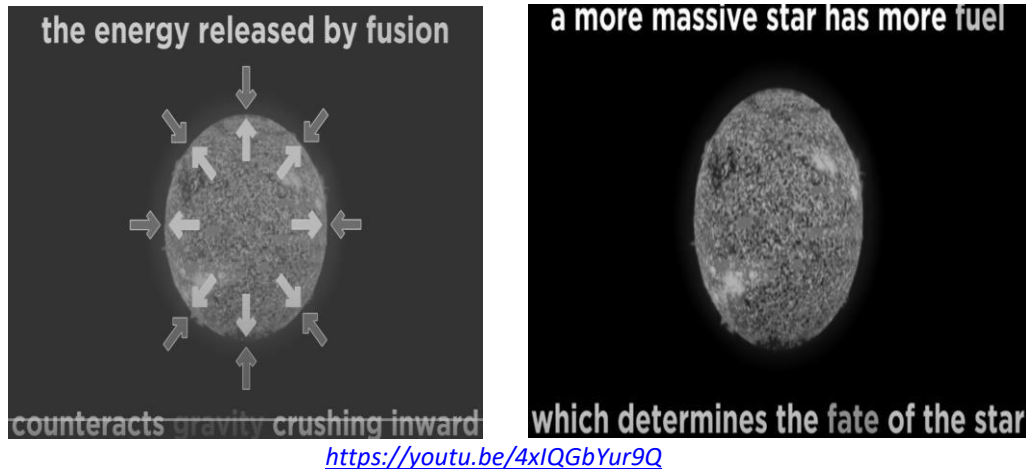


<https://youtu.be/4xIQGbYur9Q>

$$E = mc^2$$

energy mass squared
speed of light (constant)

Therefore, only by colliding nuclei together and fusing them in its ultra-hot core can a star release enough outward energy to counter the effects of gravity relentlessly crushing inward.



This means that the amount of matter that forms the star determines the amount of fuel, and through a variety of other factors, the lifetime and eventual fate of the star.

LOW MASS OR AVERAGE STAR

Given that mass is the key factor here, let's start with a low mass star.

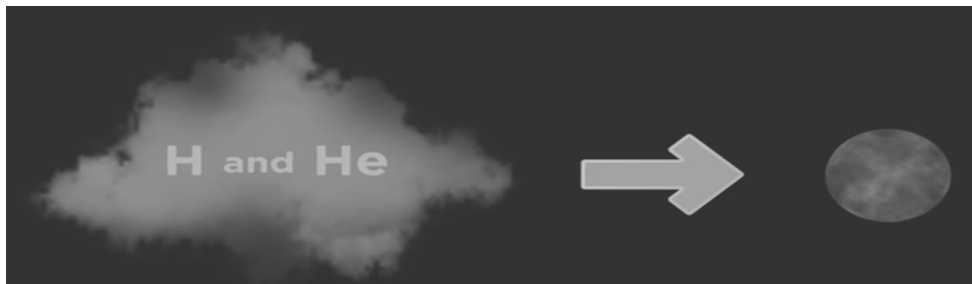
This would range from the smallest that stars can be, meaning the smallest amount of material that can sufficiently trigger nuclear fusion to qualify as a star, which is about thirteen Jupiter masses, to a star somewhere in the ballpark of our sun's mass.

As we already know, any star will begin as a cloud of gas and dust at least a few light years across.



<https://www.youtube.com/watch?v=4xIQGbYur9Q>

In the earliest era of star formation, this material was almost exclusively **hydrogen** and **helium**, as this was the remains after the brief seventeen minutes of **nucleosynthesis** soon after the Big Bang.



<https://www.youtube.com/watch?v=4xIQGbYur9Q>

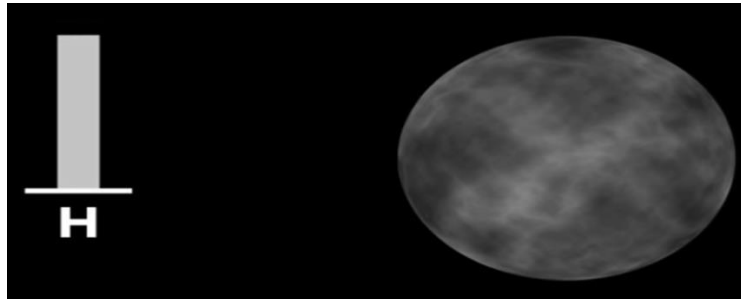
This matter collects due to gravity, pushing increasingly inward as it contracts, until things get so hot over a few millions of years. That nuclear fusion eventually begins, establishing equilibrium and generating a yellow or red main sequence star that glows with all energy released from the collision happening inside.



<https://www.youtube.com/watch?v=4xIQGbYur9Q>

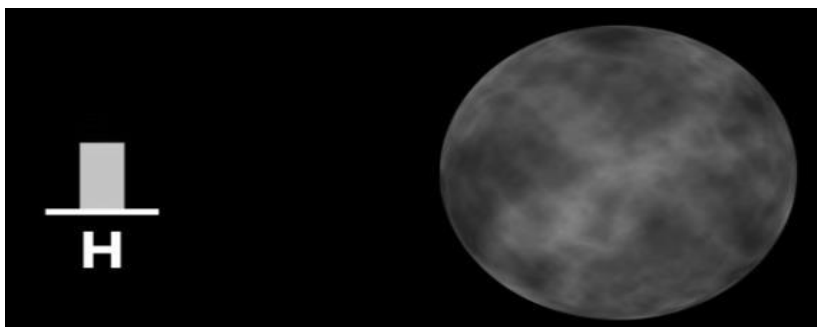
This fusion reaction begins with two protons fusing, followed by subsequent beta decay, to get proton and neutron, and we call this **deuteron**, which is a nucleus of heavy hydrogen

Then deuterons are involved in reactions that make helium which has two protons and two neutrons.



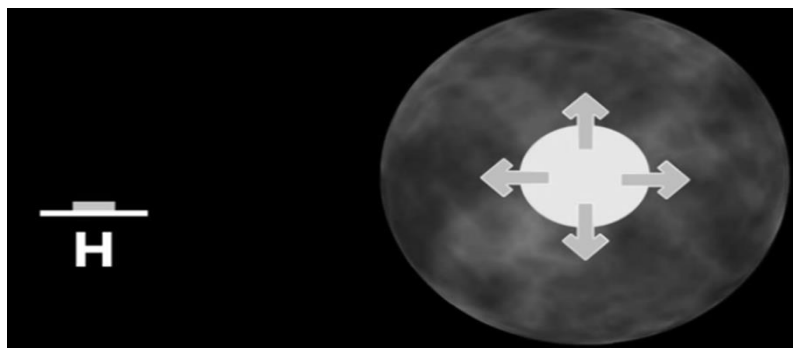
<https://www.youtube.com/watch?v=4xIQGbYur9Q>

Such a star will continue in this manner for a billions of years, slowly fusing all of the hydrogen in its core into helium and maintaining a relatively the same size, temperature and luminosity as it does so, until almost all the hydrogen is gone. At this point, things really begin to change.



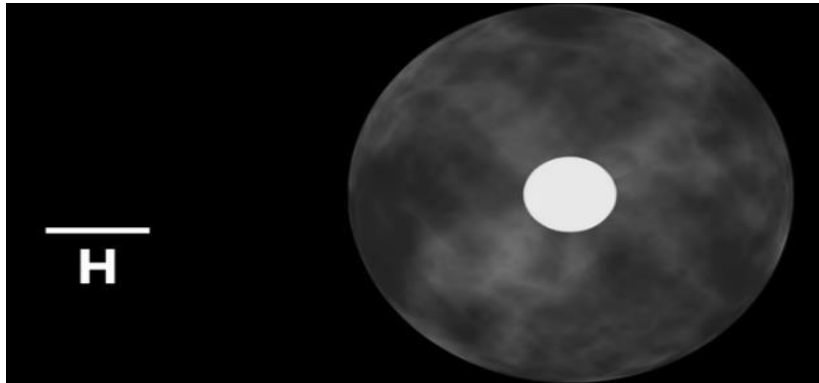
<https://www.youtube.com/watch?v=4xIQGbYur9Q>

The core of the star will shrink and get hotter, which make theremaining hydrogen burn even faster and all that extra energy being generated will be radiated outward and push the outer layer away from the core.



<https://www.youtube.com/watch?v=4xIQGbYur9Q>

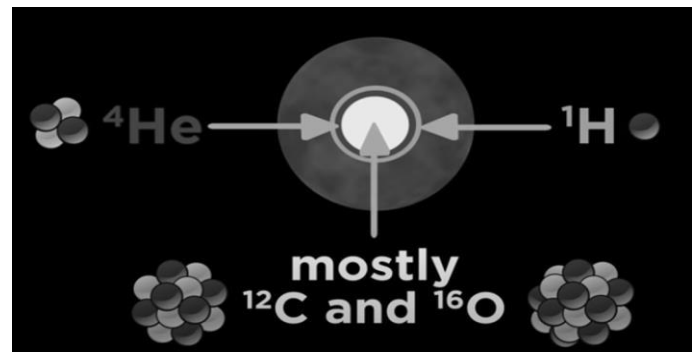
As the outer layers expand, they cool and thus become more and more red, and the star become of what we called the **red giant star**. The star can maintain this new status for a little while longer around a billion years but after almost all hydrogen is gone the core even gets smaller and even hotter.



<https://www.youtube.com/watch?v=4xIQGbYur9Q>

At this stage a phase called **helium flash** things are so hot that the stars are able to fuse this heavier helium nuclei into larger nuclei like carbon and oxygen through something called the triple alpha process and this means that the star has whole new source of fuel in all the helium it has been for billions of years.

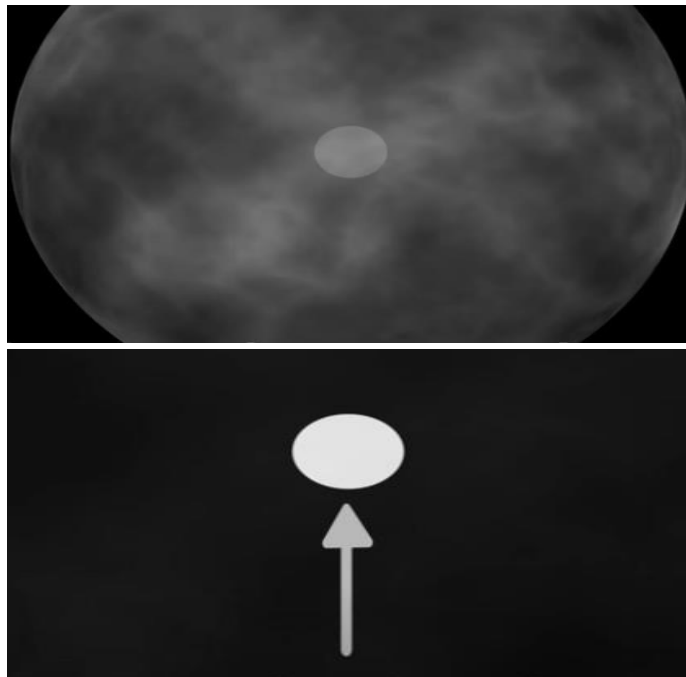
The stars begin pulsating as it runs through its final energy reserve entering what we call the horizontal branch and in this time it becomes smaller, hotter and bluer until at last much of the helium has been fused into larger nuclei.



<https://www.youtube.com/watch?v=4xIQGbYur9Q>

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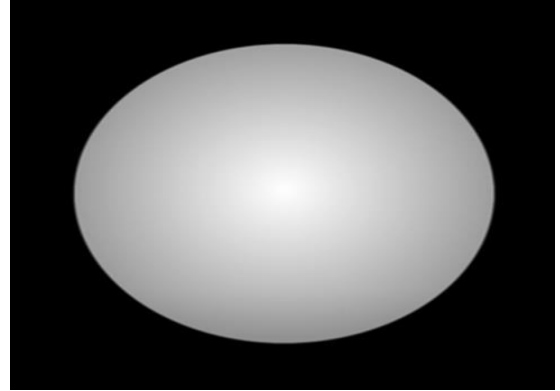
Once the core is predominantly carbon and oxygen with just a shell of helium around it and a shell of hydrogen around that, the star has very few materials left to burn, so the core will collapse and the star enters the asymptotic giant branch. This means it will grow rapidly and become a giant star again, until the last burst of energy ejects the outer layer pushing it away from the core and back into interstellar medium, leaving only a tiny very hot bare core behind about the size of the Earth.



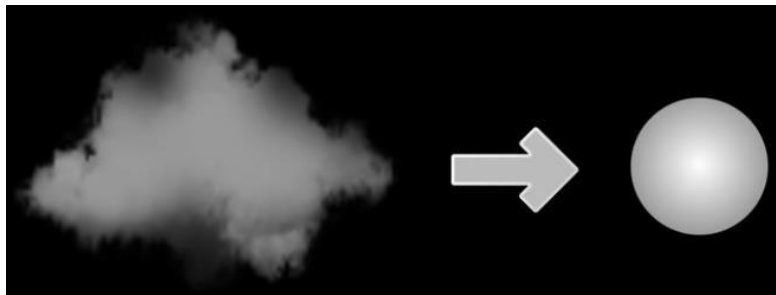
This will gradually cool as it has no more fuel to burn not being enough to fuse carbon or oxygen nuclei and it will contract further until we are left with a white dwarf star. The material in these white dwarf star will then become available to join more gas particle to form yet another star.

HIGH MASS STAR

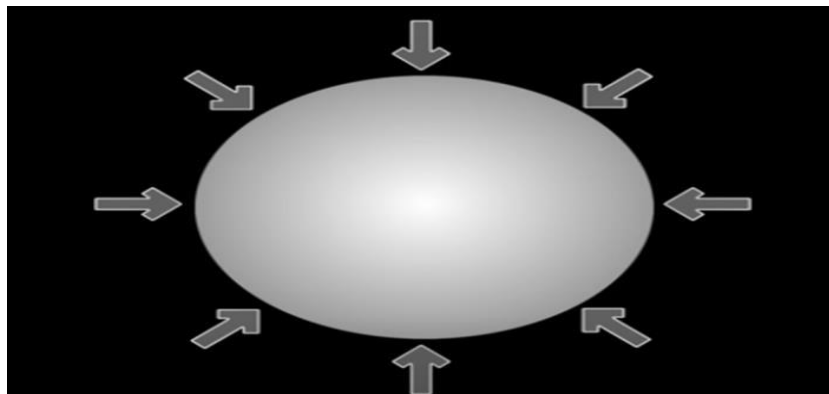
Now for a high mass star, one that is much more massive than our sun, things are quite different. Their demise will not be so quiet. Big star goes out with a bang!



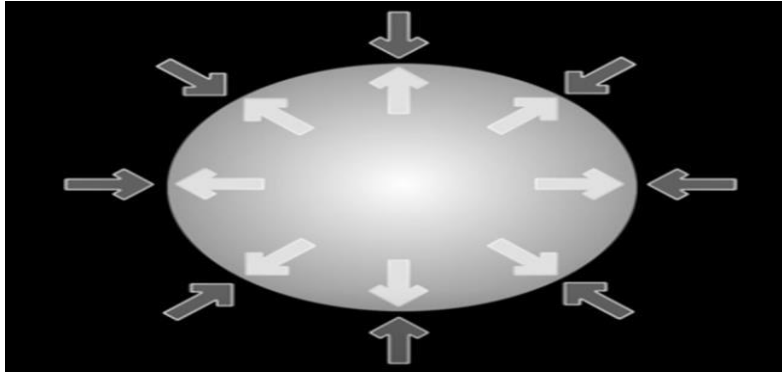
Things start out normally with a gas cloud collecting under the influence of gravity. Simply, it means that this cloud will be much larger than those that form a low mass stars, so it will contain much more mass.



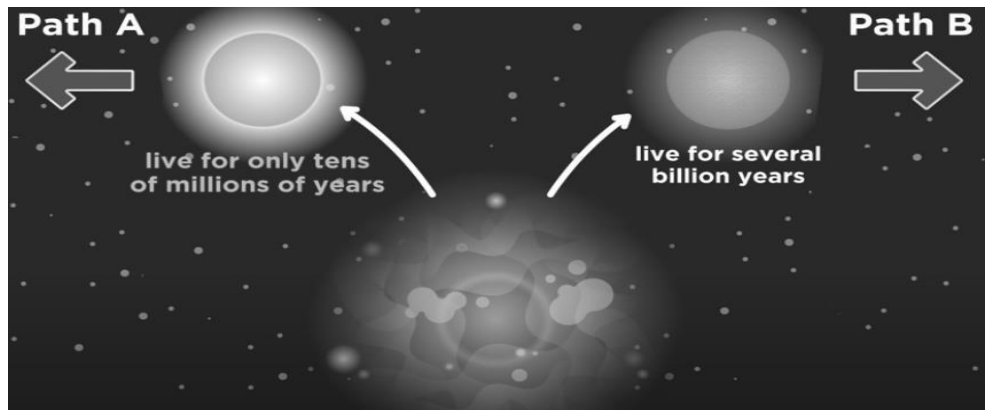
More mass means more gravity, which means the force pushing inward is much stronger and star gets much hotter.



A hotter temperature means faster fusion which generates greater outward pressure to counteract the greater inward pull of gravity.

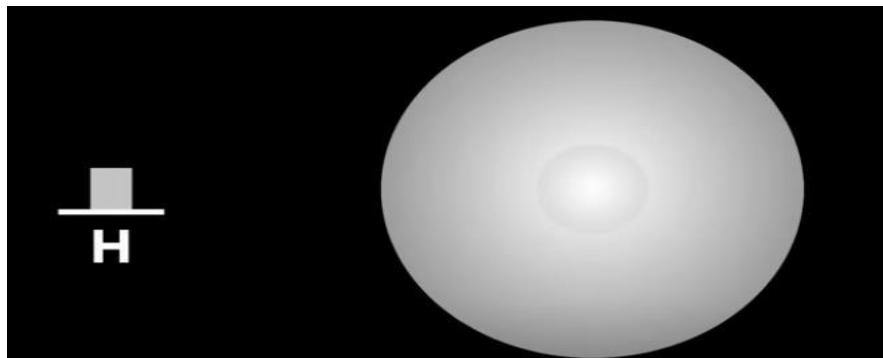


This will result in a main sequence of star that is hot, big, bright, and blue. This is where things start to go differently from low mass stars. Low mass stars take billions of years to use up all their fuel, high mass stars are much hotter and burn their fuel much faster. That means they use up all the hydrogen in their cores a hundred million years only.



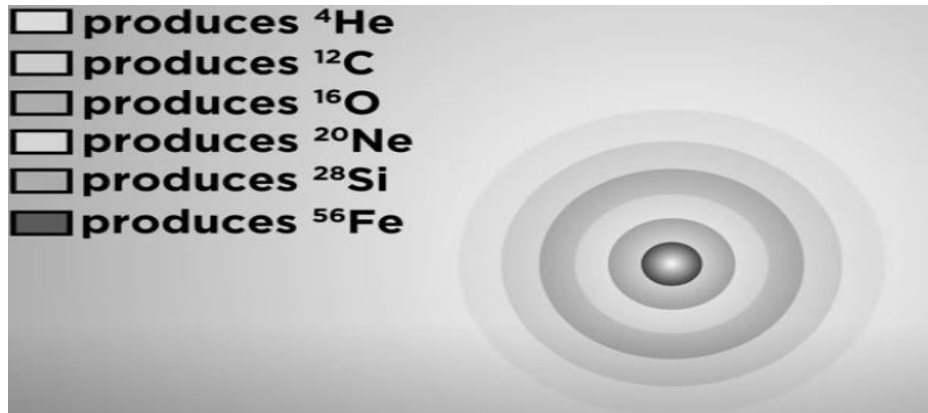
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As the fuel starts running out, the core contracts and heats up producing more energy so the star will swell up into a giant star just like we saw in a low mass stars.

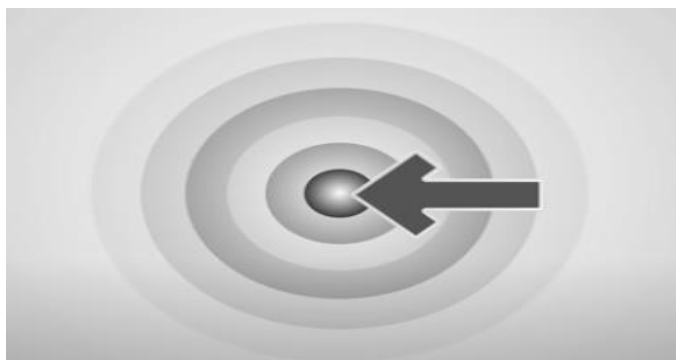


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But while the core of a high mass star continues to compress, it gets much hotter than the core of a low mass star and it becomes able to fuse helium nuclei to form **carbon** and then **oxygen, neon, silicon** each heavier nucleus being relegated to a smaller and smaller region of the core that is hot enough to fuse it. All the way at the center sits the heaviest element that can be fused within a star, **iron**.

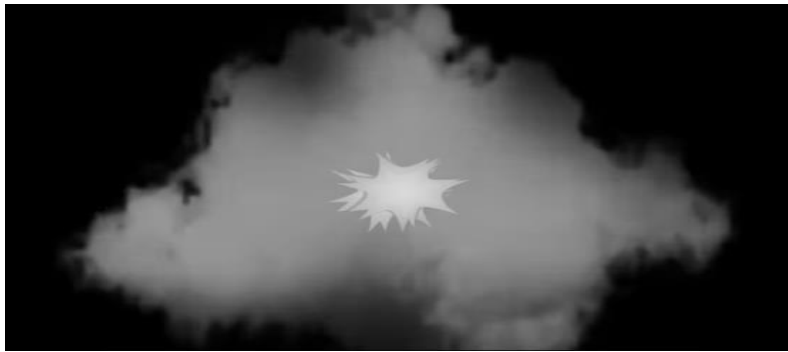


As these occurs in the different layers, each performing a particular type of fusion until no fuel remains, the star is left with a core of iron nuclei that are so stable that further fusion can release no more energy.

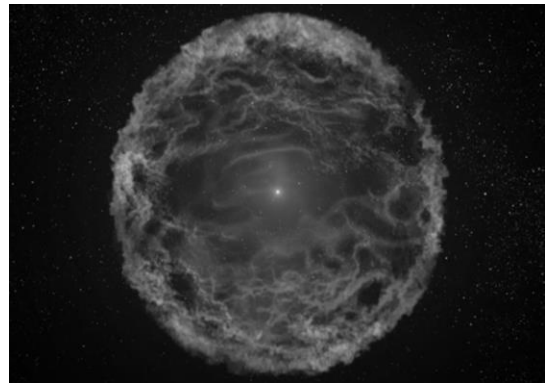


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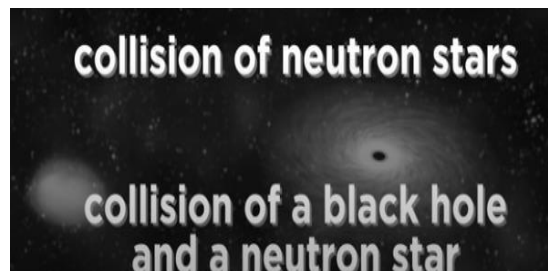
At this point gravity wins the fight and the star collapses within a second. The outer layers bouncing off the core and triggering an explosion thus ejecting all of the heavy nuclei the star has created out in space.



This awesome event, one of the violent and energetic phenomena in the universe is called a **supernova**. A supernova generates such an unbelievable burst of energy. In this brief moment dozens of elements heavier than iron can also be synthesized such as **nickel**,



copper, zinc, silver, and gold. Any element with an atomic number greater than twenty-six is made either in a supernova or a rare event like a collision of a two neutron star or a neutron star collide with black hole.

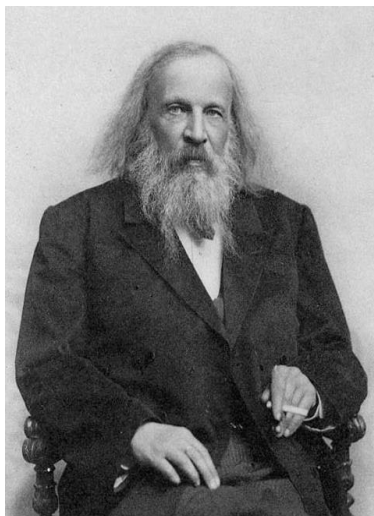


That is why these heavy element is so rare compared to elements like carbon and oxygen because stars cannot synthesized them the way they can synthesize all the elements up to iron throughout their long lives. Nature only makes these rare elements during the death of a high mass star, or in certain exotic collision events.

SYNTHESIS OF ELEMENTS

Let us have a review first. Remember that each element has a distinctive spectrum unlike any other element, you can identify the element in a light source by analyzing the light through a spectroscope and looking for characteristic patterns. Thus, the atomic spectra can help us identify elements like fingerprints to identify people. The spectrum serves as a characteristic property (something like a fingerprint) for each element, which allows scientists to identify the elements present in a sample. An isotope of an element has the same number of protons but a different number of neutrons in the nucleus of the atom.

Dmitri Mendeleev created a classification of elements based on their atomic weight. He found that organizing the elements at the time by their calculated weight demonstrated a periodic pattern of both physical and chemical properties, such as luster, physical state, reactivity to water, and others.



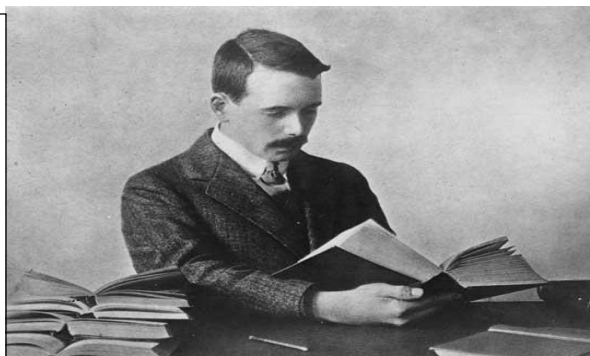
media/File:Dmitri_Mendeleev_1890s.jpg

Dmitri Ivanovich Mendeleev (often romanized as **Mendeleyev** or **Mendeleef**) was a Russian chemist and inventor. He is best remembered for formulating the Periodic Law and creating a farsighted version of the periodic table of elements. He used the Periodic Law not only to correct the then-accepted properties of some known elements, such as the valence and atomic weight of uranium, but also to predict the properties of eight elements that were yet to be discovered.

https://en.wikipedia.org/wiki/Dmitri_Mendeleev#/

Henry Moseley who was a researcher at Rutherford's laboratory. In 1913 Moseley used Rutherford's work to advance the understanding of the elements and solve the problem with Mendeleev's periodic table

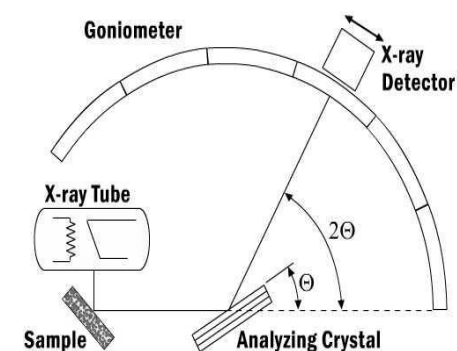
Henry Gwyn Jeffreys Moseley was an English physicist, whose contribution to the science of physics was the justification from physical laws of the previous empirical and chemical concept of the atomic number. This stemmed from his development of Moseley's law in X-ray spectra.



[https://en.wikipedia.org/wiki/Henry_Moseley#/media/File:Henry_Moseley_\(1887-1915\).jpg](https://en.wikipedia.org/wiki/Henry_Moseley#/media/File:Henry_Moseley_(1887-1915).jpg)

Organizing the elements by their weight did not always give a periodic alignment of their chemical properties. Moseley noticed that shooting electrons at elements caused them to release x-rays at unique frequencies. He also noticed that the frequency increased by a certain amount when the “positive charge” of the chosen element was higher. By arranging the elements according to the square root of the frequency they emitted, he was able to draw out an arrangement of elements that more correctly predicted periodic trends.

The experimental evidence he gave to an existing hypothesis: that the elements’ atomic number, or place in the periodic table, was uniquely tied to their “positive charge”, or the number of protons they had. This discovery allowed for a better arrangement of the periodic table, and predicted elements that were not yet discovered. His method of identifying elements by shooting electrons and looking at x-rays became a very useful tool in characterizing elements, and is now called **x-ray spectroscopy**.



https://en.wikipedia.org/wiki/X-ray_spectroscopy#/media/File:X-ray_spectroscopy_Goniometer.jpg

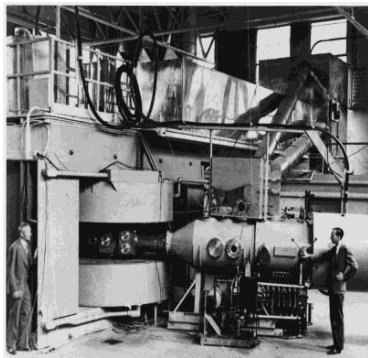
X-ray spectroscopy is a general term for several spectroscopic techniques for characterization of materials by using x-ray excitation. An example, X-ray spectrograph consists of a high voltage power supply (50 kV or 100 kV), a broad band X-ray tube, usually with a tungsten anode and a beryllium window, a specimen holder, an analyzing crystal, a goniometer, and an X-ray detector device.

[media/File:X-ray_spectroscopy_Goniometer.jpg](https://en.wikipedia.org/wiki/X-ray_spectroscopy#/media/File:X-ray_spectroscopy_Goniometer.jpg)

Synthesis of Elements

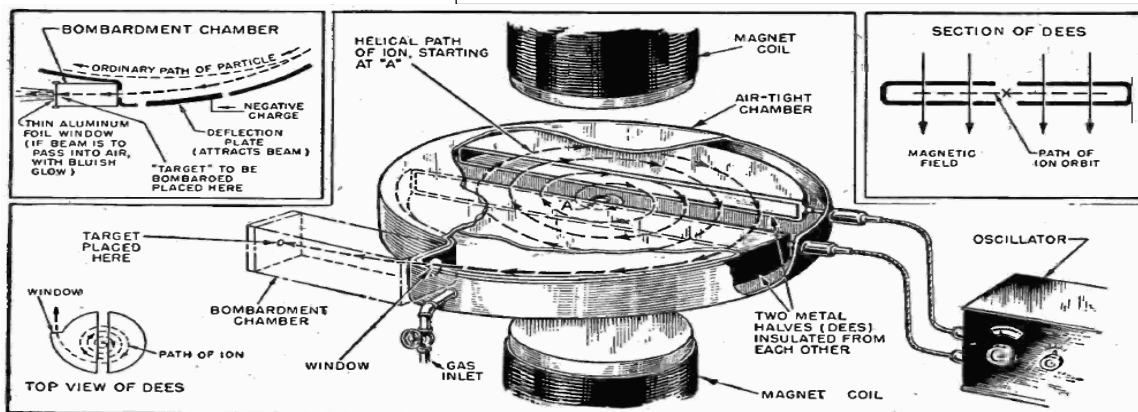
The invention of the device called **cyclotron** paved the way for

transmuting one element into another artificially. The high-energy particles that are produced from the cyclotron upon hitting heavy target nuclei produce heavier nuclei.



<https://en.wikipedia.org/wiki/Cyclotron>

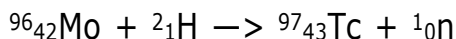
Lawrence's 60-inch cyclotron, with magnet poles 60 inches (5 feet, 1.5 meters) in diameter, at the **University of California Lawrence Radiation Laboratory**, Berkeley, in August, 1939, the most powerful accelerator in the world at the time. **Glenn T. Seaborg** and **Edwin M. McMillan** (right) used it to discover **plutonium**, **neptunium**, and many other transuranic elements and isotopes, for which they received the 1951 **Nobel Prize** in chemistry. The cyclotron's huge magnet is at left, with the flat accelerating chamber between its poles in the center. The **beamline** which analyzed the particles is at right.



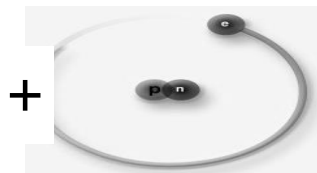
<https://en.wikipedia.org/wiki/Cyclotron>

Above is a diagram showing how a cyclotron works. The magnet's pole pieces are shown smaller than in reality; they must actually be as wide as the dees to create a uniform field.

The bombarding of **Molybdenum** with **deuteron** formed **technetium** which is the first artificially made element. Its name is derived from the Greek word **technetos** which means artificial.



Molybdenum



Deuteron

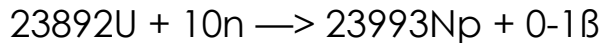


Technetium

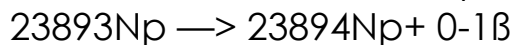
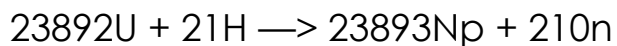
The Transuranic Elements

Transuranic elements are synthetic elements with atomic numbers higher than that of Uranium ($Z = 92$).

Neptunium ($Z = 93$) – synthesized by E.M. MacMillan in 1940



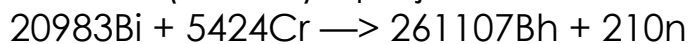
Plutonium ($Z = 94$)



The Superheavy Elements

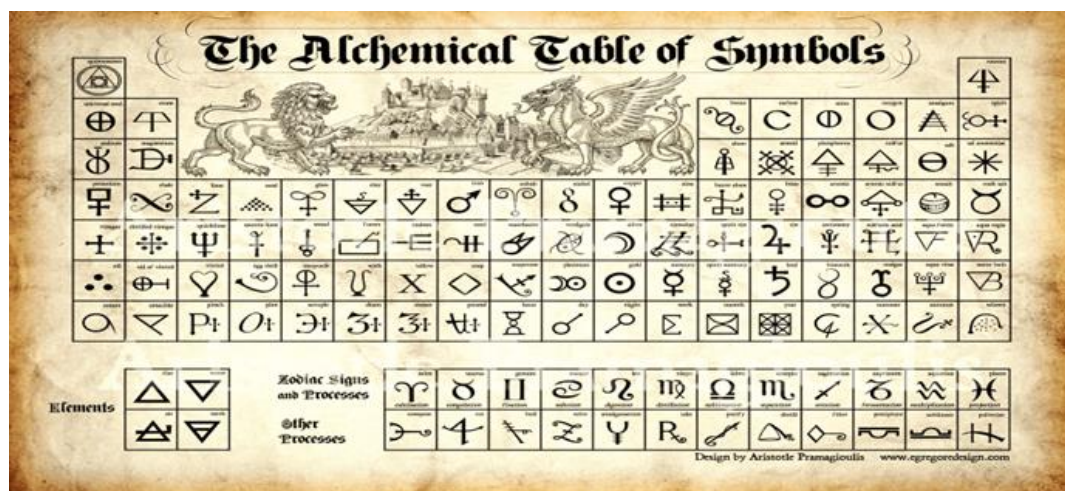
Superheavy elements are elements with atomic numbers beyond 103. These are produced by bombarding heavy nuclear targets with accelerated heavy projectiles.

Bohrium ($Z = 107$) – projectile used was Cr



Alchemy

Alchemy is the very old study and philosophy of how to change basic substances (such as metals) into other substances. It also studied how substances (and how they are changed into other substances) were related to magic and astrology. People who studied alchemy were called alchemists. Basically, it is a form of chemistry and speculative philosophy practiced in the Middle Ages and the Renaissance and concerned principally with discovering methods for transmuting baser metals into gold and with finding a universal solvent and an elixir of life.



The Alchemical Table of Symbols

https://medium.com/@Ayahuasca_yage/lectures-on-alchemy-with-terence-mckenna-e7a81430aae7

POST ACTIVITY

Essay:

Directions: Write your answer in your activity notebook. Do not write anything on this module.

Question:

If you are a star, what will be the path you want to choose? A low mass star or a heavy mass star? Why? Explain your answer in not less than 5 sentences.

III. WHAT I HAVE LEARNED

EVALUATION/POST-TEST

I. Multiple choice. Choose the answer correctly. Write the letter of the correct answers in your activity notebook. Do not write anything on this module.

1. Which of the following is a stellar core formed when the fragments of a collapsed molecular cloud contract?
 - A. protostar
 - B. supernova
 - C. red giant
 - D. main sequence star
2. The formation of a star starts with the dense regions of molecular clouds. What force pulls matter together to form these regions?
 - A. magnetic force
 - B. nuclear force
 - C. electromagnetic force
 - D. gravitational force
3. What happens when most of the hydrogen in the core is fused into helium in the stellar core?
 - A. Hydrogen fusion stops, and the pressure in the core decreases.
 - B. Hydrogen fusion continues, and the pressure in the core increases.
 - C. Gravity squeezes the star until helium and hydrogen burning occur.
 - D. Nuclear energy increases until carbon and helium burning occur.

4. Arrange the following stages of stellar evolution of a low-mass star.
- A. Protostar> main sequence star> red giant> white dwarf
 - B. Protostar> main sequence star> white dwarf >red giant>
 - C. Main sequence star> red giant> white dwarf> protostar
 - D. Main sequence star> protostar> red giant> white dwarf
5. Which of the following is the major factor predicting the fate of a star?
- A. strength of gravitational force
 - B. mass of the star
 - C. amount of iron produced
 - D. temperature of the star

II. Identification. Write the answers in your activity notebook.

6-7. The two elements formed during the creation of a low mass star.

8-10. Name at least 3 heavy elements that were formed during the formation of the heavy mass star.

III. Essay Writing

Directions: Write a short essay (maximum of 5 sentences) for each question. Do this in your notebook. Do not write anything on this learning material.

1. Explain why the atomic number is called the “fingerprint” of elements.
2. How would you relate alchemy to synthesis of new elements?
3. How did the concept of atomic number led to the synthesis of new elements?
4. Can you create elements? Support your answer.

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Bigael, Jerome. **"Formation of Light and Heavy Elements"**. Slideshare
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Teaching Guide for Senior High School **PHYSICAL SCIENCE CORE SUB**



**DEPARTMENT OF EDUCATION
SCHOOLS DIVISION OF NEGROS ORIENTAL**



SENEN PRISCILLO P. PAULIN, CESO V

Schools Division Superintendent

JOELYZA M. ARCILLA EdD

OIC - Assistant Schools Division Superintendent

MARCELO K. PALISPIS EdD JD

OIC - Assistant Schools Division Superintendent

NILITA L. RAGAY EdD

OIC - Assistant Schools Division Superintendent / CID Chief

ROSELA R. ABIERA

Education Program Supervisor – (LRMS)

ARNOLD R. JUNGCO

PSDS – Division Science Coordinator

MARICEL S. RASID

Librarian II (LRMDS)

ELMAR L. CABRERA

PDO II (LRMDS)

CRAYDON T. ABENDAN & HENDELBERT P. BARNIDO

Writers

Lay-out Artist

ALPHA QA TEAM

LIEZEL A. AGOR

EUFRADES G. ANSOK

JOAN Y. BUBULI

MA. OFELIA BUSCATO

LIELIN A. DE LA ZERNA

DEXTER D. PAIRA

BETA QA TEAM

LIEZEL AGOR BESAS

JOAN BUBULI VALENCIA

LIELIN A. DE LA ZERNA

PETER PAUL A. PATRON

THOMAS JOGIE U. TOLEDO

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SYNOPSIS

This Self Learning Kit talks about the beginning of life of a star, its journey and how it ends. Also, this kit tackles the elements that can be found along its voyage in the wide massive universe.

The synthesis of an element is the artificial creation of an element, as they cannot be found in nature. Synthesized elements are created by bombarding a known element into another element with a particle accelerator. With this learning kit, students can now explain how the concept of atomic number led to the synthesis of new elements.



Writer: **CRAYDON T. ABENDAN**, finished his Bachelor of Science in Nursing at University of Negros Occidental-Recoletos and finished his Bachelor of Science in Secondary Education major in General Science at Central Negros College, San Carlos City and just passed his CAR-MAED major in General Biology at Central Philippine State University. He is currently teaching at Jose B. Cardenas Memorial High School Senior High School Department. He is also a Grade 11 Coordinator and District Health Coordinator.



Writer: **HENDELBERT P. BARNIDO**, a graduate of Bachelor of Science in Nursing at University of Negros Occidental-Recoletos. Earned his Bachelor in Secondary units at St. Joseph College of Canlaon, Inc. on the year 2015. Currently assigned as a teacher at Jose B. Cardenas Memorial High School-Main Campus SHS. Designated as the Grade 11 HUMSS C adviser, Culture and Arts coordinator, P.E. coordinator and Dance Club adviser. Pursues a Master's Degree at Central Philippines State University.

ANSWERS KEY

Pre activity:

1. Answers may vary.

Post activity

1. Answers may vary.

Evaluation/Post-test:

- I.
 - 1.D
 - 2.D
 - 3.D
 - 4.C
 - 5.B
- II.
 6. hydrogen
 7. helium
 8. carbon
 9. silicon
 10. iron

PRE-ACTIVITY

The answer is PERIODIC PROPERTY.

What have I learned? ESSAY (answers may vary)

1. The atomic number assigned to each of the 118 elements (to date) is equal to the number of protons of each of them. This number of protons or the atomic number doesn't vary for each element and so it is its identity or "atom-print". Once the proton/s is/are compromised for each element it will not be the same element.
2. Alchemy give way to the creation and invention of new elements. the knowledge brought by alchemy is therefore used as the basis of creating new elements, especially metals.
3. All elements with atomic numbers 1 to 118 have been observed. Synthesis of new elements is accomplished by bombarding target atoms of heavy elements with ions, such that the sum of the atomic numbers of the target and ion elements equals the atomic number of the element being created.
4. You cannot create new elements by mixing different compounds. To create a new element, you have to change the number of protons in a nucleus. It is possible to do this, but it requires bombarding various ELEMENTS, one with the other, by means of high energy particle accelerators.